

EXPLORATORY DATA ANALYSIS:

Understanding the dataset to explore how the data is present in the database and if there is a need of creating some aggregated tables that can help with:

- Vendor selection for profitability
- Product pricing optimization

In []:

```
# To print the tables present in the database.
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="inventory")

cur = db.cursor()

query= """show tables from inventory"""
cur.execute(query)
data=cur.fetchall()
data
df=pd.DataFrame(data)
df
```

In []:

```
# To print the number of rows present in each table:

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector
from sqlalchemy import create_engine
from urllib.parse import quote_plus
password=quote_plus("PASSWORD@629")
engine = create_engine(f"mysql+pymysql://root:{password}@localhost:3306/inventory")
db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="inventory")

cur = db.cursor()
cur.execute("SHOW TABLES;")
tables=cur.fetchall()
for table in tables:
    table_name=table[0]
    print("-"*50,f'{table}',"-"*50)
    query= f"""select count(*) as count from {table_name}"""
    count=pd.read_sql(query,engine)['count'].values[0]
    print("Count of records:",count)
    display(pd.read_sql(f"select * from {table_name} limit 5",engine))
```

In [36]:

```
# To extract the information about a particular vendor:
import mysql.connector
from sqlalchemy import create_engine
from urllib.parse import quote_plus
password=quote_plus("PASSWORD@629")
engine = create_engine(f"mysql+pymysql://root:{password}@localhost:3306/inventory")
db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="inventory")

cur = db.cursor()
# We created this list just because the vendor number in some tables was written as VendorNumber
columns_to_check=["VendorNumber","VendorNo"]
# Looping across the tables to print the data
for table in tables:
    table_name=table[0]
    cur.execute(f"""select column_name from information_schema.columns where table_schema='{table_name}'
    and column_name in ({','.join([f'"{col}"' for col in columns_to_check])})""")
    result=cur.fetchall()
# Selecting only those tables which contain the given column.
if result:
    col_to_use = result[0][0]
    print("-"*50,f'{table}',"-"*50)
    query= f"""select count(*) as count from {table_name} WHERE {col_to_use}=4466"""
    count=pd.read_sql(query,engine)['count'].values[0]
    print("Count of records:",count)
    display(pd.read_sql(f"select * from {table_name} WHERE {col_to_use}=4466",engine))
```

----- ('purchase_prices',) -----

Count of records: 3

	Brand	Description	Price	Size	Volume	Classification	PurchasePrice	VendorNumber	VendorName
0	5215	TGI Fridays Long Island Iced	12.99	1750mL	1750	1	9.41	4466	AMERICAN VINTAGE BEVERAGE
1	5255	TGI Fridays Ultimate Mudslide	12.99	1750mL	1750	1	9.35	4466	AMERICAN VINTAGE BEVERAGE
2	3140	TGI Fridays Orange Dream	14.99	1750mL	1750	1	11.19	4466	AMERICAN VINTAGE BEVERAGE

----- ('purchases',) -----

Count of records: 2192

	InventoryId	Store	Brand	Description	Size	VendorNumber	VendorName	PONum
0	30_CULCHETH_5255	30	5255	TGI Fridays Ultimate Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE	81
1	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	81
2	1_HARDERSFIELD_5255	1	5255	TGI Fridays Ultimate Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE	81

	InventoryId	Store	Brand	Description	Size	VendorNumber	VendorName	PONum
3	38_GOULCREST_5215	38	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	82
4	59_CLAETHORPES_5215	59	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	82
...	...	...	...	...	...	...	...	...
2187	81_PEMBROKE_5215	81	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	135
2188	62_KILMARNOCK_5255	62	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE	135
2189	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	135
2190	6_GOULCREST_5215	6	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	135
2191	35_HALIVAARA_5255	35	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE	135

2192 rows × 16 columns

----- ('sales',) -----

Count of records: 9453

	InventoryId	Store	Brand	Description	Size	SalesQuantity	SalesDollars	SalesPrice
0	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
1	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
2	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
3	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
4	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
...	...	...	...	...	...	...	...	...
9448	9_BLACKPOOL_5215	9	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
9449	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte	1.75L	1	12.99	12.99

	InventoryId	Store	Brand	Description	Size	SalesQuantity	SalesDollars	SalesPrice
				Mudslide				
9450	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1	12.99	12.99
9451	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1	12.99	12.99
9452	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1	12.99	12.99

9453 rows × 14 columns

----- ('vendor_invoice',) -----

Count of records: 55

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity	Dollars	Freight
0	4466	AMERICAN VINTAGE BEVERAGE	2024-01-07	8137	2023-12-22	2024-02-21	15	140.55	8.5
1	4466	AMERICAN VINTAGE BEVERAGE	2024-01-19	8207	2023-12-27	2024-02-26	335	3142.33	16.9
2	4466	AMERICAN VINTAGE BEVERAGE	2024-01-18	8307	2024-01-03	2024-02-18	41	383.35	1.9
3	4466	AMERICAN VINTAGE BEVERAGE	2024-01-27	8469	2024-01-14	2024-03-11	72	673.20	3.3
4	4466	AMERICAN VINTAGE BEVERAGE	2024-02-04	8532	2024-01-19	2024-03-15	79	740.21	3.4
5	4466	AMERICAN VINTAGE BEVERAGE	2024-02-09	8604	2024-01-24	2024-03-15	347	3261.37	17.6
6	4466	AMERICAN VINTAGE BEVERAGE	2024-02-17	8793	2024-02-05	2024-04-02	72	675.36	3.1
7	4466	AMERICAN VINTAGE BEVERAGE	2024-03-01	8892	2024-02-12	2024-03-28	117	1096.05	5.1
8	4466	AMERICAN VINTAGE BEVERAGE	2024-03-07	8995	2024-02-19	2024-04-02	129	1209.27	5.4
9	4466	AMERICAN VINTAGE BEVERAGE	2024-03-12	9033	2024-02-22	2024-04-16	147	1377.87	6.6
10	4466	AMERICAN VINTAGE BEVERAGE	2024-03-16	9180	2024-03-03	2024-04-19	211	1979.33	9.5

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity	Dollars	Freight
11	4466	AMERICAN VINTAGE BEVERAGE	2024-03-23	9244	2024-03-08	2024-04-21	161	1510.69	6.9
12	4466	AMERICAN VINTAGE BEVERAGE	2024-03-31	9371	2024-03-17	2024-05-13	176	1649.20	8.9
13	4466	AMERICAN VINTAGE BEVERAGE	2024-04-09	9491	2024-03-24	2024-05-08	215	2016.43	10.0
14	4466	AMERICAN VINTAGE BEVERAGE	2024-04-17	9583	2024-03-31	2024-05-12	110	1035.10	5.6
15	4466	AMERICAN VINTAGE BEVERAGE	2024-04-20	9639	2024-04-04	2024-06-04	515	5323.01	25.0
16	4466	AMERICAN VINTAGE BEVERAGE	2024-04-29	9800	2024-04-15	2024-06-07	275	2775.01	13.6
17	4466	AMERICAN VINTAGE BEVERAGE	2024-05-09	9886	2024-04-21	2024-06-12	312	3050.86	13.7
18	4466	AMERICAN VINTAGE BEVERAGE	2024-05-14	9999	2024-04-29	2024-06-26	310	3117.82	14.9
19	4466	AMERICAN VINTAGE BEVERAGE	2024-05-16	10095	2024-05-06	2024-06-27	215	2123.01	11.4
20	4466	AMERICAN VINTAGE BEVERAGE	2024-05-28	10169	2024-05-11	2024-07-04	327	3298.55	15.8
21	4466	AMERICAN VINTAGE BEVERAGE	2024-06-04	10257	2024-05-17	2024-07-08	376	3708.18	20.3
22	4466	AMERICAN VINTAGE BEVERAGE	2024-06-12	10346	2024-05-23	2024-07-20	640	6458.38	29.7
23	4466	AMERICAN VINTAGE BEVERAGE	2024-06-17	10445	2024-05-30	2024-07-19	288	2937.56	15.8
24	4466	AMERICAN VINTAGE BEVERAGE	2024-06-22	10600	2024-06-09	2024-08-01	308	3213.28	17.0
25	4466	AMERICAN VINTAGE BEVERAGE	2024-06-29	10695	2024-06-16	2024-08-12	143	1585.75	8.0
26	4466	AMERICAN VINTAGE BEVERAGE	2024-07-09	10777	2024-06-22	2024-08-15	18	179.88	0.9
27	4466	AMERICAN VINTAGE	2024-07-08	10836	2024-06-25	2024-08-14	8	85.96	0.4

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity	Dollars	Freight
		BEVERAGE							
28	4466	AMERICAN VINTAGE BEVERAGE	2024-07-11	10969	2024-06-29	2024-08-18	94	993.12	5.0
29	4466	AMERICAN VINTAGE BEVERAGE	2024-07-19	11085	2024-07-06	2024-09-04	601	5883.09	31.1
30	4466	AMERICAN VINTAGE BEVERAGE	2024-07-26	11187	2024-07-14	2024-08-26	1535	15406.69	77.0
31	4466	AMERICAN VINTAGE BEVERAGE	2024-08-03	11244	2024-07-18	2024-09-04	266	2888.22	15.3
32	4466	AMERICAN VINTAGE BEVERAGE	2024-08-11	11362	2024-07-26	2024-09-24	206	2171.64	11.0
33	4466	AMERICAN VINTAGE BEVERAGE	2024-08-18	11489	2024-08-04	2024-09-16	768	7292.98	37.9
34	4466	AMERICAN VINTAGE BEVERAGE	2024-08-24	11540	2024-08-08	2024-10-02	1207	12242.43	64.8
35	4466	AMERICAN VINTAGE BEVERAGE	2024-09-02	11716	2024-08-19	2024-09-29	433	4334.43	20.3
36	4466	AMERICAN VINTAGE BEVERAGE	2024-09-12	11771	2024-08-23	2024-10-11	370	3640.18	18.2
37	4466	AMERICAN VINTAGE BEVERAGE	2024-09-20	11901	2024-09-01	2024-10-30	358	3626.74	16.3
38	4466	AMERICAN VINTAGE BEVERAGE	2024-09-25	11993	2024-09-07	2024-10-23	233	2264.85	12.2
39	4466	AMERICAN VINTAGE BEVERAGE	2024-10-01	12125	2024-09-16	2024-11-07	284	2813.46	15.4
40	4466	AMERICAN VINTAGE BEVERAGE	2024-10-08	12235	2024-09-23	2024-11-20	258	2592.08	11.9
41	4466	AMERICAN VINTAGE BEVERAGE	2024-10-09	12253	2024-09-23	2024-11-14	1	11.19	0.0
42	4466	AMERICAN VINTAGE BEVERAGE	2024-10-12	12321	2024-09-26	2024-11-19	172	1685.98	7.5
43	4466	AMERICAN VINTAGE BEVERAGE	2024-10-20	12466	2024-10-05	2024-11-26	280	2798.60	14.5

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity	Dollars	Freight
44	4466	AMERICAN VINTAGE BEVERAGE	2024-10-27	12515	2024-10-09	2024-11-30	178	1763.28	8.6
45	4466	AMERICAN VINTAGE BEVERAGE	2024-11-07	12702	2024-10-21	2024-12-11	183	1812.77	8.7
46	4466	AMERICAN VINTAGE BEVERAGE	2024-11-12	12752	2024-10-25	2024-12-11	216	2156.16	10.5
47	4466	AMERICAN VINTAGE BEVERAGE	2024-11-20	12828	2024-10-30	2024-12-18	262	2597.36	12.9
48	4466	AMERICAN VINTAGE BEVERAGE	2024-11-27	12929	2024-11-06	2025-01-04	270	2622.32	12.8
49	4466	AMERICAN VINTAGE BEVERAGE	2024-11-28	13092	2024-11-16	2024-12-30	209	2036.17	11.2
50	4466	AMERICAN VINTAGE BEVERAGE	2024-12-06	13134	2024-11-20	2025-01-18	305	3018.89	15.7
51	4466	AMERICAN VINTAGE BEVERAGE	2024-12-16	13254	2024-11-28	2025-01-13	262	2530.36	11.3
52	4466	AMERICAN VINTAGE BEVERAGE	2024-12-26	13432	2024-12-09	2025-01-27	231	2263.43	11.3
53	4466	AMERICAN VINTAGE BEVERAGE	2024-12-30	13483	2024-12-13	2025-02-11	221	2178.27	11.3
54	4466	AMERICAN VINTAGE BEVERAGE	2025-01-09	13627	2024-12-22	2025-02-05	413	3985.03	19.5

In [13]:

```
import mysql.connector
from sqlalchemy import create_engine
from urllib.parse import quote_plus
password=quote_plus("PASSWORD@629")
engine = create_engine(f"mysql+pymysql://root:{password}@localhost:3306/inventory")
db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="inventory")

import pandas as pd
purchases=pd.read_sql_query("select * from purchases where VendorNumber=4466",engine)
sales=pd.read_sql_query("select * from sales where VendorNo=4466",engine)
purchase_price=pd.read_sql_query("select * from purchase_prices where VendorNumber=4466",engine)
vendor_invoice=pd.read_sql_query("select * from vendor_invoice where VendorNumber=4466",engine)
display(purchases)
display(sales)
```

	InventoryId	Store	Brand	Description	Size	VendorNumber	VendorName	PONumber
0	30_CULCHETH_5255	30	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE	81
1	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	81
2	1_HARDERSFIELD_5255	1	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE	81
3	38_GOULCREST_5215	38	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	82
4	59_CLAETHORPES_5215	59	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	82
...	...	...	...	...	...	...	...	...
2187	81_PEMBROKE_5215	81	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	135
2188	62_KILMARNOCK_5255	62	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE	135
2189	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	135
2190	6_GOULCREST_5215	6	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE	135
2191	35_HALIVAARA_5255	35	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE	135

2192 rows × 16 columns

	InventoryId	Store	Brand	Description	Size	SalesQuantity	SalesDollars	SalesPrice
0	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
1	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
2	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
3	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
4	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99

	InventoryId	Store	Brand	Description	Size	SalesQuantity	SalesDollars	SalesPrice
...	...	...	...	...	...	...	...	...
9448	9_BLACKPOOL_5215	9	5215	TGI Fridays Long Island Iced	1.75L	1	12.99	12.99
9449	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1	12.99	12.99
9450	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1	12.99	12.99
9451	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1	12.99	12.99
9452	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1	12.99	12.99

9453 rows × 14 columns

In [5]:

```
purchases.groupby(['Brand', 'PurchasePrice'])[['Quantity', 'Dollars']].sum()
```

Out[5]:

		Quantity	Dollars
Brand	PurchasePrice		
3140	11.19	4640	51921.60
5215	9.41	4923	46325.43
5255	9.35	6215	58110.25

In [20]:

```
sales.groupby(['Brand', 'SalesPrice'])[['SalesQuantity', 'SalesDollars']].sum()
```

Out[20]:

		SalesQuantity	SalesDollars
Brand	SalesPrice		
3140	12.99	3890	50531.10
5215	12.99	4651	60416.49
5255	12.99	6096	79187.04

- The purchase table contains the actual purchase data, including the date of purchase, brands purchased by vendors, the amount paid in dollars and the quantity purchased
- The purchase price column is derived from the purchase_prices table which provides product wise actual and purchase prices. The combination of vendor and brand id unique in this table
- The vendor_invoice table aggregates the data from purchases table, summarizing quantity and dollars amount, along with and additional column for freight. This table maintains uniqueness based on vendor and POnumber.

- The sales table captures actual sales transactions, detailing the brands purchased by vendors, the quantity sold, the selling price and the revenue earned.

As the data we need for analysis is distributed in different tables, we need to create a summary table containing:

- Purchase transactions made by vendors
- Sales transaction data
- Freight cost for each vendor
- actual product prices from vendors

Purchase Transactions

In [14]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="inventory")

cur = db.cursor()

query= """ select p.VendorNumber,p.VendorName,p.Brand,pr.Volume,p.PurchasePrice,pr.Price
from inventory.purchase_prices as pr join inventory.purchases as p
on pr.Brand=p.Brand
where p.PurchasePrice>0

group by p.VendorNumber,p.Brand,p.VendorName,p.PurchasePrice, pr.Price,pr.Volume
order by sum(Dollars);"""
cur.execute(query)
data=cur.fetchall()
df=pd.DataFrame(data,columns=["VendorNumber","VendorName","Brand","Volume","PurchasePrice","ActualPrice","TotalQuantity","Amount"])
display(df)
```

	VendorNumber	VendorName	Brand	Volume	PurchasePrice	ActualPrice	TotalQuantity	Amount
0	7245	PROXIMO SPIRITS INC.	3065	50	0.71	0.99	1	
1	3960	DIAGEO NORTH AMERICA INC	6127	200	1.47	1.99	1	
2	3924	HEAVEN HILL DISTILLERIES	9123	50	0.74	0.99	2	
3	8004	SAZERAC CO INC	5683	50	0.39	0.49	6	
4	9815	WINE GROUP INC	8527	750	1.32	4.99	2	
...	...	...	...	...	...	...	...	
10687	3960	DIAGEO NORTH	3545	1750	21.89	29.99	138109	302

	VendorNumber	VendorName	Brand	Volume	PurchasePrice	ActualPrice	TotalQuantity	Amou
		AMERICA INC						
10688	3960	DIAGEO NORTH AMERICA INC	4261	1750	16.17	22.99	201682	326
10689	17035	PERNOD RICARD USA	8068	1750	18.24	24.99	187407	341
10690	4425	MARTIGNETTI COMPANIES	3405	1750	23.19	28.99	164038	380
10691	1128	BROWN- FORMAN CORP	1233	1750	26.27	36.99	145080	381

10692 rows × 8 columns

Sales Transactions

In [6]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="inventory")

cur = db.cursor()

query= """ select VendorNo,VendorName,Brand,SalesPrice,sum(ExciseTax),sum(SalesQuantity)
from inventory.sales
group by VendorNo,VendorName,Brand,SalesPrice
order by sum(SalesDollars);"""
cur.execute(query)
data=cur.fetchall()
df=pd.DataFrame(data,columns=["VendorNumber","VendorName","Brand","PricePerProduct","Tot
display(df)
```

	VendorNumber	VendorName	Brand	PricePerProduct	TotalExcise	TotalQuantity	AmountEarne
0	9552	M S WALKER INC	19669	0.00	2.67	24	0.0
1	3960	DIAGEO NORTH AMERICA INC	3046	0.00	77.22	42	0.0
2	8892	TRINCHERO FAMILY ESTATES	3963	0.00	6.31	8	0.0
3	2000	SOUTHERN WINE & SPIRITS NE	19465	0.00	3.24	29	0.0

	VendorNumber	VendorName	Brand	PricePerProduct	TotalExcise	TotalQuantity	AmountEarne
4	4425	MARTIGNETTI COMPANIES	25340	0.00	2.25	20	0.0
...	...	...	...	...	...	...	...
19960	3960	DIAGEO NORTH AMERICA INC	4261	22.99	155840.91	84815	1949896.8
19961	4425	MARTIGNETTI COMPANIES	3405	29.99	148371.48	80750	2421692.5
19962	17035	PERNOD RICARD USA	8068	23.49	193945.34	105552	2479416.4
19963	3960	DIAGEO NORTH AMERICA INC	3545	29.99	152475.87	82984	2488690.1
19964	1128	BROWN-FORMAN CORP	1233	34.99	150037.64	81657	2857178.4

19965 rows × 7 columns

In [1]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="inventory")

cur = db.cursor()

query= """ select VendorNumber,VendorName,round(sum(Freight),2) as Total_Freight from in
group by VendorNumber,VendorName
order by VendorNumber;"""
cur.execute(query)
data=cur.fetchall()
df=pd.DataFrame(data,columns=["VendorNumber","VendorName","Freight"])
display(df)
```

	VendorNumber	VendorName	Freight
0	2	IRA GOLDMAN AND WILLIAMS, LLP	27.08
1	54	AAPER ALCOHOL & CHEMICAL CO	0.48
2	60	ADAMBA IMPORTS INTL INC	367.52
3	105	ALTAMAR BRANDS LLC	62.39
4	200	AMERICAN SPIRITS EXCHANGE	6.19
...	...	...	...
124	98450	Serralles Usa LLC	856.02
125	99166	STARK BREWING COMPANY	130.09

	VendorNumber	VendorName	Freight
126	172662	SWEETWATER FARM	178.34
127	173357	TAMWORTH DISTILLING	202.50
128	201359	FLAVOR ESSENCE INC	0.09

129 rows × 3 columns

Overall vendor performance

In [28]:

```
import time
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import mysql.connector

db=mysql.connector.connect( host="localhost",
                           username="root",
                           password="PASSWORD@629",
                           database="inventory")

cur = db.cursor()
start = time.time()
query= """ with pur as (select p.VendorNumber,p.VendorName,p.Brand,pr.Volume,p.PurchaseP
sum(Dollars) as TotalAmountSpent
from inventory.purchase_prices as pr join inventory.purchases as p
on pr.Brand=p.Brand
where p.PurchasePrice>0
group by p.VendorNumber,p.Brand,p.VendorName,p.PurchasePrice, pr.Price,pr.Volume
order by sum(Dollars)),

sls as(select VendorNo,VendorName,Brand,SalesPrice,sum(ExciseTax)as TotalExcise
,sum(SalesQuantity) as TotalQuantitySold,sum(SalesDollars) as TotalEarned
from inventory.sales
group by VendorNo,VendorName,Brand,SalesPrice
order by sum(SalesDollars)),

frg as(select VendorNumber,VendorName,round(sum(Freight),2) as Total_Freight from invent
group by VendorNumber,VendorName
order by VendorNumber)

select pur.VendorNumber,pur.VendorName,pur.Brand,pur.Price as ActualPrice,pur.PurchasePr
pur.TotalQuantityPurchased,pur.TotalAmountSpent,pur.Volume,
sls.SalesPrice as SalesPricePerQuantity,sls.TotalQuantitySold,sls.TotalEarned,
sls.TotalExcise,frg.Total_Freight
from
pur join sls
on pur.VendorNumber=sls.VendorNo
and pur.Brand=sls.Brand
join frg
on pur.VendorNumber=frg.VendorNumber
"""

cur.execute(query)
data=cur.fetchall()
```

```

vendor_sales_summary=pd.DataFrame(data,columns=["VendorNumber","VendorName","Brand","Act
"PurPricePerProduct","TotalQuantityPurchased","TotalAmount
"SalesPricePerQuantity","TotalQuantitySold","TotalEarned","
"TotalFreight"])
display(vendor_sales_summary)
end = time.time()

```

	VendorNumber	VendorName	Brand	ActualPricePerProduct	PurPricePerProduct	TotalQuantityF
0	9552	M S WALKER INC	19669	11.99	6.06	
1	3960	DIAGEO NORTH AMERICA INC	3046	47.99	35.81	
2	8892	TRINCHERO FAMILY ESTATES	3963	14.99	11.36	
3	2000	SOUTHERN WINE & SPIRITS NE	19465	6.99	4.72	
4	2000	SOUTHERN WINE & SPIRITS NE	19465	6.99	4.72	
...	...	...	...	...	...	
22281	4425	MARTIGNETTI COMPANIES	3405	28.99	23.19	
22282	4425	MARTIGNETTI COMPANIES	3405	28.99	23.19	
22283	17035	PERNOD RICARD USA	8068	24.99	18.24	
22284	3960	DIAGEO NORTH AMERICA INC	3545	29.99	21.89	
22285	1128	BROWN- FORMAN CORP	1233	36.99	26.27	

22286 rows × 13 columns

This query generates a vendor-wise sales and purchase summary, which is valuable for:

Performance optimization:

- This query involves heavy joins and aggregations on large datasets like sales and purchases
- Storing pre-aggregated results avoids repeated expensive computations
- Helps in analyzing, purchases and pricing for different vendors and brands
- Future benefits for storing this data for faster dashboarding and reporting
- Instead of running expensive queries each time, dashboards can fetch data quickly from vendor_sales_summary

Now, we will be cleaning our final table

i) Checking the datatypes in each column

In [31]:

```
vendor_sales_summary.dtypes
```

Out[31]:

```
VendorNumber      int64
VendorName        object
Brand             int64
ActualPricePerProduct  float64
PurPricePerProduct  float64
TotalQuantityPurchased  object
TotalAmountSpent    float64
Volume            object
SalesPricePerQuantity  float64
TotalQuantitySold    object
TotalEarned        float64
TotalExcise        float64
TotalFreight       float64
dtype: object
```

ii) Checking if there are null values if yes, then we'll replace them with zero

In [32]:

```
vendor_sales_summary.isnull().sum()
```

Out[32]:

```
VendorNumber      0
VendorName        0
Brand             0
ActualPricePerProduct  0
PurPricePerProduct  0
TotalQuantityPurchased  0
TotalAmountSpent    0
Volume            0
SalesPricePerQuantity  0
TotalQuantitySold    0
TotalEarned        0
TotalExcise        0
TotalFreight       0
dtype: int64
```

iii) Checking for white spaces in the string values

In [33]:

```
vendor_sales_summary["VendorName"].unique()
```

Out[33]:

```
array(['M S WALKER INC', 'DIAGEO NORTH AMERICA INC',  
      'TRINCHERO FAMILY ESTATES', 'SOUTHERN WINE & SPIRITS NE',  
      'HEAVEN HILL DISTILLERIES', 'ULTRA BEVERAGE COMPANY LLP',  
      'CONSTELLATION BRANDS INC', 'MARTIGNETTI COMPANIES',  
      'STE MICHELLE WINE ESTATES', 'BACARDI USA INC',  
      'WINE GROUP INC', 'MILTONS DISTRIBUTING CO',  
      'KOBRA BRAND CORPORATION', 'THE IMPORTED GRAPE LLC',  
      'STATE WINE & SPIRITS', 'E & J GALLO WINERY',  
      'WEIN BAUER INC', 'DELICATO VINEYARDS INC',  
      'JIM BEAM BRANDS COMPANY', 'PINE STATE TRADING CO',  
      'SAZERAC NORTH AMERICA INC.', 'VINEYARD BRANDS INC',  
      'Circa Wines', 'SAZERAC CO INC',  
      'CRUSH WINES', 'TY KU LLC',  
      'SURVILLE ENTERPRISES CORP', 'DIAGEO CHATEAU ESTATE WINES',  
      'PALM BAY INTERNATIONAL INC', 'BANFI PRODUCTS CORP',  
      'PSP WINES', 'PERNOD RICARD USA',  
      'SHAW ROSS INT L IMP LTD', 'FREDERICK WILDMAN & SONS',  
      'BRONCO WINE COMPANY', 'VINILANDIA USA',  
      'HIGHLAND WINE MERCHANTS LLC', 'FORTUNE WINE BROKERS LLC',  
      'PERFECTA WINES', 'Dunn Wine Brokers',  
      'TRUETT HURST', 'LOYAL DOG WINERY',  
      'VINEXTRA INC', 'PROXIMO SPIRITS INC.',  
      'REMY COINTREAU USA INC', 'TREASURY WINE ESTATES',  
      'THE PIERPONT GROUP LLC', 'MOET HENNESSY USA INC',  
      'LATITUDE BEVERAGE COMPANY', 'DUGGANS DISTILLED PRODUCTS',  
      'MHW LTD', 'BROWN-FORMAN CORP',  
      'DJINN SPIRITS LLC', 'UNCORKED',  
      'LAIRD & CO', 'R.P.IMPORTS INC',  
      'STELLAR IMPORTING CO LLC', 'SOUTHERN GLAZERS W&S OF NE',  
      'Serralles Usa LLC', 'FLAG HILL WINERY & VINEYARD',  
      'Russian Standard Vodka', 'BULLY BOY DISTILLERS',  
      'LUXCO INC', 'CENTEUR IMPORTS LLC',  
      'CASTLE BRANDS CORP.', 'CHARLES JACQUIN ET CIE INC',  
      'STOLI GROUP,(USA) LLC', 'WILLIAM GRANT & SONS INC',  
      'BLACK PRINCE DISTILLERY INC', 'APPOLO VINEYARDS LLC',  
      'SEA HAGG DISTILLERY LLC', 'STAR INDUSTRIES INC.',  
      'FABRIZIA SPIRITS LLC', 'PHILLIPS PRODUCTS CO.',  
      'OLE SMOKY DISTILLERY LLC', 'MAJESTIC FINE WINES',  
      'WESTERN SPIRITS BEVERAGE CO', 'DISARONNO INTERNATIONAL LLC',  
      'VRANKEN AMERICA', 'CAMPARI AMERICA',  
      'PARK STREET IMPORTS LLC', 'TAMWORTH DISTILLING',  
      'MCCORMICK DISTILLING CO', 'AMERICAN SPIRITS EXCHANGE',  
      'CAPSTONE INTERNATIONAL', 'KLIN SPIRITS LLC',  
      'SEA BREEZE CELLARS LLC', 'FANTASY FINE WINES CORP',  
      'SILVER MOUNTAIN CIDERS', 'ALTAMAR BRANDS LLC',  
      'EDRINGTON AMERICAS', 'JEWELL TOWNE VINEYARDS',  
      'HOOD RIVER DISTILLERS, Inc.',  
      'IRA GOLDMAN AND WILLIAMS, LLP',  
      'POVERTY LANE ORCHARDS', 'ATLANTIC IMPORTING COMPANY',  
      'ADAMBA IMPORTS INTL INC', 'VINEYARD BRANDS LLC',  
      'CALEDONIA SPIRITS INC', 'MARTIGNETTI COMPANIES',  
      'ALISA CARR BEVERAGES', 'MARSALLE COMPANY',  
      'FLAVOR ESSENCE INC', 'MOONLIGHT MEADERY',  
      'GILMANTON WINERY & VINEYARD', 'PREMIER DISTRIBUTORS',  
      'NICHE W & S', 'BLACK ROCK SPIRITS LLC',  
      'VINEDREA WINES LLC', 'SWEETWATER FARM',  
      'WALPOLE MTN VIEW WINERY', 'HAUNTING WHISPER VYDS',  
      'LABELLE VYDS AND WINERY', 'MANGO BOTTLING INC',
```



```
'INCREDIBREW INC', 'BLACK COVE BEVERAGES',
'CANDIA VINEYARDS', 'FULCHINO VINEYARD INC',
'TAKARA SAKE USA INC', 'SIDNEY FRANK IMPORTING CO',
'SMOKY QUARTZ DISTILLERY LLC', 'TALL SHIP DISTILLERY LLC',
'ZORVINO VINEYARDS', 'STARK BREWING COMPANY',
'PREMIUM PORT WINES, INC.', 'SWEET BABY VINEYARD',
'AMERICAN VINTAGE BEVERAGE'], dtype=object)
```

In [35]:

```
vendor_sales_summary['Volume']=vendor_sales_summary['Volume'].astype('float64')
```

In [36]:

```
vendor_sales_summary['TotalQuantitySold']=vendor_sales_summary['TotalQuantitySold'].ast
```

In [37]:

```
vendor_sales_summary['TotalQuantityPurchased']=vendor_sales_summary['TotalQuantityPurcha
```

In [38]:

```
vendor_sales_summary['VendorName']=vendor_sales_summary['VendorName'].str.strip()
```

Now, after removing the inconsistencies, we will be creating a few more columns that would help us analyze the data better

In [39]:

```
vendor_sales_summary['GrossProfit']=vendor_sales_summary['TotalEarned']-vendor_sales_sum
```

In [40]:

```
vendor_sales_summary['ProfitMargin']=(vendor_sales_summary['GrossProfit']/vendor_sales_s
```

In [41]:

```
vendor_sales_summary['TotalQuantityPurchased'].dtype
```

Out[41]:

```
dtype('int64')
```

In [42]:

```
vendor_sales_summary['StockTurnOver']=round(vendor_sales_summary['TotalQuantitySold']/ve
```

In [43]:

```
vendor_sales_summary['SalesToPurchaseRatio']=vendor_sales_summary['TotalEarned']/vendor_
```

Now, we will be saving this table into our database

In [44]:

```
df=vendor_sales_summary
display(df)
```

	VendorNumber	VendorName	Brand	ActualPricePerProduct	PurPricePerProduct	TotalQuantityF
0	9552	M S WALKER INC	19669	11.99	6.06	

	VendorNumber	VendorName	Brand	ActualPricePerProduct	PurPricePerProduct	TotalQuantityF
1	3960	DIAGEO NORTH AMERICA INC	3046	47.99	35.81	
2	8892	TRINCHERO FAMILY ESTATES	3963	14.99	11.36	
3	2000	SOUTHERN WINE & SPIRITS NE	19465	6.99	4.72	
4	2000	SOUTHERN WINE & SPIRITS NE	19465	6.99	4.72	
...	...	...	...	...	...	
22281	4425	MARTIGNETTI COMPANIES	3405	28.99	23.19	
22282	4425	MARTIGNETTI COMPANIES	3405	28.99	23.19	
22283	17035	PERNOD RICARD USA	8068	24.99	18.24	
22284	3960	DIAGEO NORTH AMERICA INC	3545	29.99	21.89	
22285	1128	BROWN- FORMAN CORP	1233	36.99	26.27	

22286 rows × 17 columns

In [45]:

```
import pandas as pd
from sqlalchemy import create_engine, text
from urllib.parse import quote_plus

password = quote_plus("PASSWORD@629")
engine = create_engine(f"mysql+pymysql://root:{password}@localhost:3306/inventory")

def ingest_db(df, table_name, engine, chunksize=50000): # write 50k rows at a time
    df.to_sql(table_name, con=engine, if_exists='replace', index=False, chunksize=chunks

ingest_db(df, "vendor_sales_summary", engine)
check = pd.read_sql("SELECT * FROM vendor_sales_summary LIMIT 5;", engine)
print(check.head())
```

	VendorNumber	VendorName	Brand	ActualPricePerProduct	\
0	9552	M S WALKER INC	19669	11.99	
1	3960	DIAGEO NORTH AMERICA INC	3046	47.99	
2	8892	TRINCHERO FAMILY ESTATES	3963	14.99	
3	2000	SOUTHERN WINE & SPIRITS NE	19465	6.99	
4	2000	SOUTHERN WINE & SPIRITS NE	19465	6.99	

	PurPricePerProduct	TotalQuantityPurchased	TotalAmountSpent	Volume	\
0	6.06	267	1618.02	750.0	
1	35.81	1995	71440.95	1750.0	
2	11.36	1947	22117.92	750.0	
3	4.72	1159	5470.48	750.0	
4	4.72	1159	5470.48	750.0	

	SalesPricePerQuantity	TotalQuantitySold	TotalEarned	TotalExcise	\
0	0.0	24	0.0	2.67	
1	0.0	42	0.0	77.22	
2	0.0	8	0.0	6.31	
3	0.0	29	0.0	3.24	
4	0.0	29	0.0	3.24	

	TotalFreight	GrossProfit	ProfitMargin	StockTurnOver	\
0	55551.82	-1618.02	-100.0	0.0899	
1	257032.07	-71440.95	-100.0	0.0211	
2	11595.97	-22117.92	-100.0	0.0041	
3	207.76	-5470.48	-100.0	0.0250	
4	18808.83	-5470.48	-100.0	0.0250	

	SalesToPurchaseRatio
0	0.0
1	0.0
2	0.0
3	0.0
4	0.0

-----Our Data Is Loaded into MySQL-----

In []: